

# Diabetes Risk Analysis Using Tableau

## Introduction

Diabetes is a chronic condition influenced by behavioral, genetic, and physiological factors. This project visualizes and interprets diabetes risk factors using Tableau's advanced analytics capabilities to make complex health data accessible to non-experts. The dashboard provides insights into patterns, risk trends, and correlations between demographic, lifestyle, and clinical variables.

## Objective

The objective of this project is to explore relationships between health indicators and diabetes risk levels through Tableau visualizations. Key goals include: - Identifying critical health risk factors (BMI, glucose, physical inactivity, etc.) - Exploring lifestyle and demographic influences - Classifying individuals into Low, Moderate, and High-Risk clusters - Differentiating preventable vs. non-preventable factors - Laying the groundwork for predictive risk modeling integration using Python (TabPy)

## Dataset Description

The dataset contains 100,000 individual patient records across 31 fields representing demographics, lifestyle, medical history, and clinical measurements. Each record includes a binary diabetes indicator (Yes/No) and a continuous diabetes risk score. Key Domains: - Demographics: Age, Gender, Education, Income - Lifestyle: Physical activity, Diet, Sleep hours, Screen time, Smoking, Alcohol - Medical History: Hypertension, Family History, Cardiovascular history - Clinical Indicators: Glucose (fasting/postprandial), HbA1c, BMI, Blood Pressure Derived Fields: Age Category, BMI Category, Activity Category, Sleep Category, Hypertension Category, and Risk Clusters (manual clustering simulation using calculated fields).

## Data Analysis and Visualization

Data was cleaned, normalized, and analyzed using correlation matrices in Python to identify the top predictors: HbA1c, Glucose Fasting, Glucose Postprandial, Diabetes Risk Score, Family History, BMI, Age, Physical Activity, and Systolic BP. Four Tableau dashboards were created: 1. Demographic Overview — visualizes age, gender, and socioeconomic impact on diabetes prevalence. 2. Lifestyle Influence — explores how diet, sleep, activity, and screen time affect risk. 3. Health History Insights — examines hypertension, family history, and cardiovascular conditions. 4. Clinical Deep Dive — shows clinical relationships like BMI vs HbA1c, glucose vs insulin, and clustering results. Example Insights: - Diet vs Diabetes Risk: Negative correlation; better diet lowers risk ( $p < 0.0001$ ) - Glucose vs Insulin: Strong positive correlation; insulin resistance observed - Cluster 1: Young & healthy | Cluster 2: Older, high BMI | Cluster 3: Moderate risk

## Predictive Model Integration (TabPy)

To enhance the interactivity and predictive value, a Random Forest model was integrated with Tableau using TabPy. Model trained on: HbA1c, Glucose Fasting, Glucose Postprandial, Family History, Age, Physical Activity, BMI, and Systolic BP. Model Performance: - Accuracy: 92% - AUC: 94% - F1 Score: 93% The model outputs a real-time diabetes risk classification (Yes/No) and confidence score directly in Tableau.

## Conclusion

This Tableau project successfully integrates exploratory analytics, statistical modeling, and predictive capabilities. The visual dashboards provide actionable insights into diabetes risk determinants across

demographics, lifestyle, medical history, and clinical domains. Combined with TabPy integration, it supports real-time health risk assessment and promotes data-driven decision-making in healthcare.

## References

- Dataset: <https://www.kaggle.com/datasets/mohankrishnathalla/diabetes-health-indicators-dataset> -  
Tableau Dashboard Inspirations: DataQuest, Tableau Public - ML References: Kaggle Notebooks on  
Diabetes Risk Prediction - TabPy Integration Tutorials: YouTube – “Integrate Python in Tableau”